Geometry Teacher Answer Key

Partner A and Partner B answers are identical for each numbered problem.

Unit 1: Basics & Reasoning

#	1.1 Segments, Midpoints & Angle Basics	1.2 Logic & Reasoning
1	12	162
2	8	19
3	22	$x = -5$
4	$(5, 6)$	False; a rectangle that is not a square
5	5	Law of Detachment
6	$45°$	Law of Syllogism
7	$x = 15$	Distributive Property; $x = 6$
8	$52°$	$110°$

Unit 2: Angles & Parallel Lines

#	2.1 Angle Pairs & Transversals	2.2 Triangles, Polygons & Parallel Lines
1	$65°$	$x = 7$
2	$115°$	$115°$
3	$x = 15$	$90°$
4	$x = 26$	$70°$
5	$x = 13$	8
6	$x = 70$	$120°$
7	$132°$	$40°$
8	$135°$	$85°$

Unit 3: Triangles & Congruence

#	3.1 Proving Triangles Congruent	3.2 Triangle Properties
1	SSS	$x = 10$
2	SAS	45
3	ASA	6
4	HL	7
5	No (AAA only gives similar triangles)	13
6	14	11
7	$48°$	12
8	7	scalene, right

Unit 4: Similarity

#	4.1 Similar Figures & Proportions	4.2 Similarity Rules, Right Triangles & Dilations
1	6	AA
2	$\frac{5}{2}$	SSS $\sim$
3	12	SAS $\sim$
4	36	8
5	45	6
6	18 ft	10
7	3	$(6, -3)$
8	18	$\frac{5}{2}$

Unit 5: Right Triangles & Trigonometry

#	5.1 Pythagorean Theorem & Special Triangles	5.2 Trigonometric Ratios
1	10	$\frac{3}{5}$
2	13	$\frac{5}{12}$
3	obtuse	$\frac{8}{17}$
4	$7\sqrt{2}$	5
5	$5\sqrt{3}$	7.7
6	$6\sqrt{2}$	13.1
7	16	$22.6°$
8	12 (ft / m)	17.3 (ft / m)

Unit 6: Circles

#	6.1 Arcs, Angles & Segments in Circles	6.2 Circumference, Area & Equations of Circles
1	$35°$	14π
2	$40°$	25π
3	2π	7
4	6π	$x^2 + y^2 = 25$
5	8	center $(3, -2),\ r = 4$
6	12	$(x + 1)^2 + (y - 4)^2 = 9$
7	$30°$	4
8	$60°$	8

Unit 7: Area, Surface Area & Volume

#	7.1 Area of Polygons & Circles	7.2 Surface Area & Volume
1	30	120
2	40	45π
3	36	12π
4	9π	36π
5	$60 - 4\pi$	150
6	$9\sqrt{3}$	42π
7	48	60
8	16π	81

Unit 8: Coordinate Geometry & Transformations

#	8.1 Coordinate Geometry	8.2 Transformations
1	13	$(6, -2)$
2	$(3, 5)$	$(5, 3)$
3	$(5, 8)$	$(7, 2)$
4	2	$(-4, 3)$
5	24	$(3, -5)$
6	20	$(6, -4)$
7	16	$(-3, 4)$
8	$(2, 1)$	90° counterclockwise